

# ZIYUE (ALVIN) LIU

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## EDUCATION

**University of California, Santa Barbara**, Santa Barbara, CA  
Doctor of Philosophy in Computer Science, GPA: 4.00/4.00

Sep. 2023 – Apr. 2027

**University of California, Santa Barbara**, Santa Barbara, CA  
Master of Arts in Statistics, GPA: 3.97/4.00

Sep. 2021 – Mar. 2023

**Nankai University — School of Mathematical Sciences**, Tianjin, China  
Bachelor of Science in Statistics, GPA: 3.72/4.00

Aug. 2016 – Jun. 2020

## RESEARCH

**Efficient and Resilient Pre-Training of Large Language Models**

Sep. 2023 – Now

- Efficient model architecture, optimizer, and infrastructure on leadership-class HPC.
- Resilient pre-training system via fault-tolerant communication primitives for O(100k+) GPUs.

## SELECTED PUBLICATIONS

- [1] Z. Liu, Z. Wang, R. Zhang, A. Maurya, H. Zhou, P. Hovland, S. Di, F. Cappello, B. Nicolae, and Z. Zhang. **"ReCoVer: Resilient LLM Pre-Training System via Fault-Tolerant Collective and Versatile Workload."** NeurIPS, 2026.
- [2] Z. Liu, R. Zhang, Z. Wang, Y. Zhao, Y. Su, Z. Yang and Z. Zhang. **"Muon<sup>2</sup>: Boosting Muon via Adaptive Second-Moment Preconditioning."** EMNLP oral, 2026.
- [3] Z. Liu\*, R. Zhang\*, Z. Wang\*, M. Yan, Z. Yang, P. Hovland, B. Nicolae, F. Cappello, S. Tang and Z. Zhang. **"CoLA: Compute-Efficient Pre-Training of LLMs via Low-Rank Activation."** EMNLP Oral, 2025. (\*Equal contribution)
- [4] Z. Wang\*, Z. Liu\*, R. Zhang, A. Maurya, B. Nicolae, P. Hovland, F. Cappello, and Z. Zhang. **"BOOST: Bottleneck-Optimized Scalable Training Framework for Low-Rank Large Language Models."** MLSys, 2026. (\*Equal contribution)
- [5] R. Zhang\*, Z. Liu\*, Z. Wang and Z. Zhang. **"LaX: Boosting Low-Rank Training of Foundation Models via Latent Crossing."** NeurIPS, 2025. (\*Equal contribution)
- [6] Z. Liu, Y. Li, J. Hu, X. Yu, S. Shiau, Xin Ai, Z. Zeng, and Z. Zhang. **"DeepOHeat: Operator Learning-based Ultra-fast Thermal Simulation in 3D-IC Design."** Design Automation Conference (DAC), 2023.
- [7] Z. Yang, Z. Liu, S. Choudhary, X. Xie, C. Gao, S. Kunzmann and Z. Zhang. **"CoMERA: Computing- and Memory-Efficient Training via Rank-Adaptive Tensor Optimization."** NeurIPS, 2024.
- [8] R. Zhang, Y. Zhao, Z. Liu, Z. Wang and Z. Zhang. **"Muon+: Towards Better Muon via One Additional Normalization Step."** Under review, 2026.
- [9] R. Zhang, Y. Zhao, Z. Liu, Z. Wang, D. Li, Y. Su, S. Liu and Z. Zhang. **"TEON: Tensorized Orthonormalization Beyond Layer-Wise Muon for Large Language Model Pre-Training."** Under review, 2026.
- [10] J. Zhou, Y. Yang, K. Zhen, Z. Liu, Y. Zhao, E. Banijamali, A. Mouchtaris, N. Wong and Z. Zhang. **"QuZO: Quantized Zeroth-Order Fine-Tuning for Large Language Models."** EMNLP, 2025.

## INTERNSHIPS

**Google Cloud**, Sunnyvale, CA

Software Engineer Intern: Efficient MoE adapter tuning and multi-tenant serving

June. 2026 – Sep. 2026

**Argonne National Laboratory**, Lemont, IL

Visiting Student: Fault-tolerant LLM pre-training

April. 2025 – Sep. 2025

**Cadence Design Systems**, Austin, TX

Research Intern: Data-driven modeling of real-world data center thermal/CFD simulations

Jun. 2024 – Sep. 2024

Research Intern: Data-driven modeling of large-scale automotive aerodynamic simulations

Apr. 2023 – Sep. 2023

Research Intern: Physics-informed operator learning for 3D-IC thermal simulations

Jun. 2022 – Sep. 2022